CISS245: Advanced Programming
Quiz q6301

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Q1. Correct the following.
Answer:

```
#include <iostream>
#include <iomanip>

class Time
{
public:
    Time(int hour, int min, int sec)
        : hour_(hour), min_(min), sec_(sec)
    {}

    void println() const
    {
        std::cout << std::setw(2) << std::setfill('0') << hour_ << ':'
                  << std::setw(2) << std::setfill('0') << min_ << ':'
                  << std::setw(2) << std::setfill('0') << sec_ << '\n';
    }

private:
    int hour_, min_, sec_;
};

int main()
{
    Time t(1, 2, 3);
    t.println();
    return 0;
}
```

Q2. Complete *and correct* the following.
Answer:

```
#include <iostream>
#include <iomanip>

class Rational
{
public:
    Rational(int n, int d)
    : n_(n), d_(d)
```

```
    {}
    Rational max(Rational r)
    {
        if(n_ * r.d_ > r.n_ * d_) return *this
        else return r;
    }
    // ... other methods not shown ...
private:
    int n_, d_; // numerator and denominator
};

int main()
{
    Rational r0(1, 2);
    Rational r1(2, 3);
    Rational r2 = r0.max(r1); // r2 models 2/3
    Rational r3 = r1.max(r0); // r3 models 2/3

    // In other words f.max(g) returns the max of f and g.
    // (Similar to the max of the Int class from my notes.)

    return 0;
}
```

Q3. Complete *and correct* the following.
Answer:

```
#include <iostream>
#include <iomanip>

class Robot
{
public:
    Robot()
        : x_(0), y_(0)
    {}
    Robot& walk_east()
    {
        x_++;
        return *this;
    }
private:
    int x_, y_;
};

int main()
{
    Robot r2d2;                             // r2d2.x_ is 0
    r2d2.walk_east();                       // r2d2.x_ is 1
    r2d2.walk_east().walk_east();           // r2d2.x_ is 3
```

```
    r2d2.walk_east().walk_east().walk_east(); // r2d2.x_ is 6
    return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.